ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 07
29-09-2024

JEE(Main+Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Review (Unit Test # 01, 02, 03, 04 & 05)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	A	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,B,C	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.50	375.00	60.00	250.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	6	8	8	2	1	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	B	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	A,B,C	B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	200.00	2.00	4.00	1.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	1	9	5	2	3	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	A	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C,D	A,B,C,D	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	25.00	3.00	3.00	19.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	1	9	2	40	2	7

HINT – SHEET

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PART-1 : PHYSICS

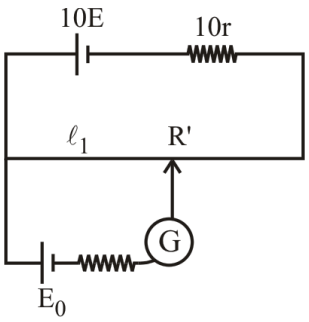
SECTION-I (i)

1. Ans (A)

In 1st case

$$\rho_1 = \frac{10E}{10r + R'} \times \frac{R'}{L}$$

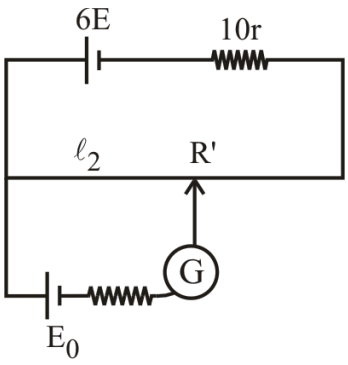
Hence $E_0 = r_1 \ell_1$



It's in 2nd case, $\rho_2 = \frac{6E}{10r + R'} \times \frac{R'}{L}$

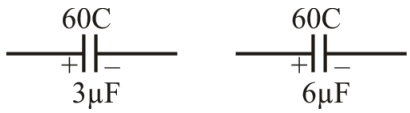
and $E_0 = r_2 \ell_2$

$$\therefore \frac{\ell_1}{\ell_2} = \frac{\rho_2}{\rho_1} = \frac{6}{10} = \frac{3}{5}$$

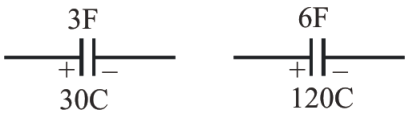


2. Ans (C)

Charge initially

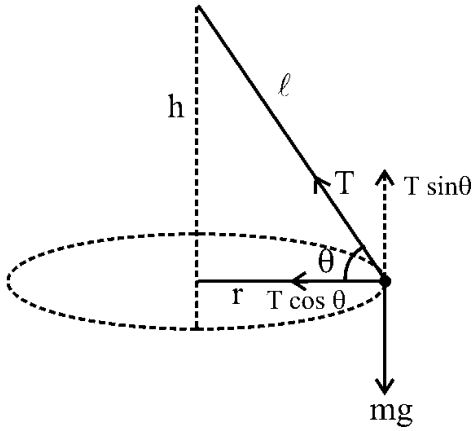


Charge finally



for this charge arrangement to happen, 90C charge has to flow upwards.

3. Ans (A)

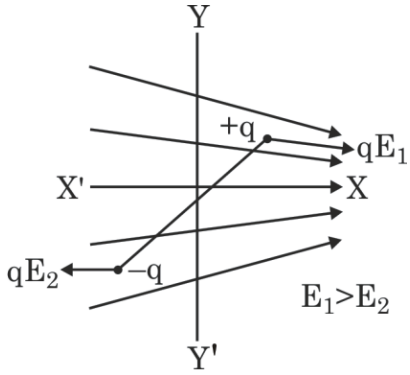


$T \sin\theta = mg$

$T \times h/\ell = mg$

$T = \frac{mg\ell}{h}$

4. Ans (A)



Electric field lines are dense

$qE_1 > qE_2$

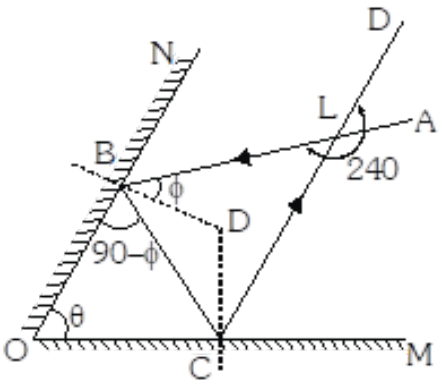
the dipole move in ⊕ve x direction ⊥ distance from force component provide Torque in clockwise direction.

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PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**



Le the mirror OM and ON make θ angle with one another.

Let Incident ray AB make ϕ angle with mirror.

$\angle BLC = 240 - 180 = 60°$

In ΔABC sum of all angles = 180°

$2\phi + 2(\theta - \phi) + 60° = 180°$

$2\phi + 2\theta - 2\phi + 60° = 180°$

$\theta = 60°$

As per the problem $\Rightarrow \theta = 60°$

No. of images = $\left(\frac{360°}{60°} - 1\right) = 5$

6. **Ans (A,B,C)**

$\sin\theta = \frac{R}{R+h}$

$\theta = \sin^{-1}\left(\frac{R}{R+h}\right)$

curved area on earth = $2\pi R^2(1 - \sin\theta)$

area on earth escaped from this satellite

$= 4\pi R^2 - 2\pi R^2(1 - \sin\theta)$

$2\pi R^2(1 + \sin\theta)$

7. **Ans (A,B,C,D)**

Since the bullet returns to its point of projection, its net displacement is zero, which is choice (a).

The bullet take 5 sec to reach the maximum height. Therefore, initial speed (u) of the bullet is :

(∵ final velocity = 0)

$$u = gt = 10 \times 5 = 50 \text{ ms}^{-1}$$

directed upwards which is choice (d).

The maximum height (h) attained by the bullet is,

$$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (5)^2 = 125 \text{ m}$$

Therefore, the total distance traveled by the bullet in 10 s = 125 + 125 = 250 m, which is choice (b).

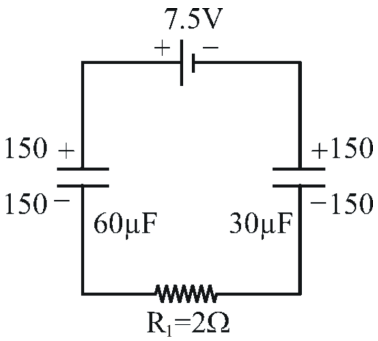
For height h << radius of earth, the magnitude of g is constant, i.e., the rate of change of velocity is constant.

PART-1 : PHYSICS

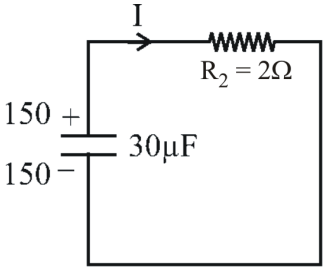
SECTION-II (i)

1. **Ans (2.50)**

Long time after closing the switch S_1



Just after closing the switch S_2



$-I \times 2 + \frac{150}{30} = 0$

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2. **Ans (375.00)**

$H = \frac{Q^2}{2C}$

$Q = 150\ \mu C$

$C = 30\ \mu F$

3. **Ans (60.00)**

Velocity is maximum at a = 0

$N = mpg$

4. **Ans (250.00)**

$$\frac{dv}{dt} = -\left(\frac{\alpha}{t_0/2}\right)t + \alpha$$

$$v = -\left(\frac{2\alpha}{t_0} \times \frac{t^2}{2}\right) + \alpha t$$

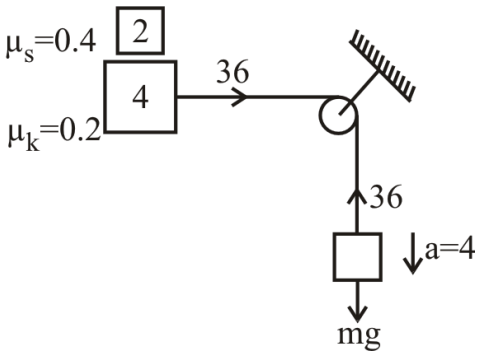
$$\frac{dx}{dt} = -\frac{\alpha}{t_0} \times t^2 + \alpha t$$

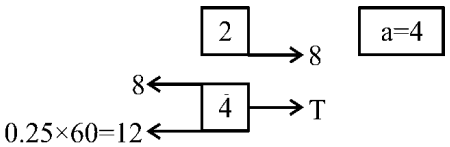
$$x = -\frac{\alpha t^3}{3t_0} + \frac{\alpha t^2}{2}$$

PART-1 : PHYSICS

SECTION-II (ii)

5. **Ans (6)**





$T - 20 = 4 \times 4$

$T = 36$

$10m - 36 = M \times 4$

$6M = 36$

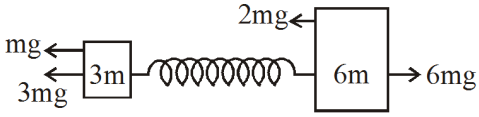
$M = 6$

6. **Ans (8)**



$a_{com} = \frac{3mg}{9m} = g/3$

from C.O.M. frame



$4mgx_1 + 4mgx_2 = \frac{1}{2}K(x_1 + x_2)^2$

$4mg = \frac{1}{2}K(x_1 + x_2)$

$x_1 + x_2 = \frac{8mg}{K}$

$n = 8$

7. **Ans (8)**

Conserving angular momentum :

$m(v_1\cos 60°)4R = mv_2R \Rightarrow \frac{v_2}{v_1} = 2.$

Conserving energy of the system :

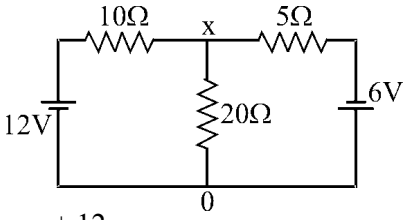
$$-\frac{GMm}{4R} + \frac{1}{2}mv_1^2 = -\frac{GMm}{R} + \frac{1}{2}mv_2^2$$

$$\Rightarrow \frac{1}{2}v_2^2 - \frac{1}{2}v_1^2 = \frac{3}{4}\frac{GM}{R}$$

$$\Rightarrow v_1^2 = \frac{1}{2}\frac{GM}{R} \Rightarrow v_1 = \frac{1}{\sqrt{2}}\sqrt{64 \times 10^6} = \frac{8000}{\sqrt{2}}\text{m/s}$$

$$\Rightarrow 500\sqrt{2}\,X = \frac{8000}{\sqrt{2}} \Rightarrow X = 8$$

8. **Ans (2)**



$$\frac{x}{20} + \frac{x+6}{5} + \frac{x+12}{10} = 0$$

$$\frac{2x + 8x + 48 + 4x - 48}{40} = 0$$

$\Rightarrow x = 0 \Rightarrow i_1 = i_2$

$i_1 = \frac{12}{10} \Rightarrow i_1 + i_2 = \frac{12}{5}$

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9. **Ans (1)**

$\tan\theta = 2t$

$\sec^2\theta\frac{d\theta}{dt} = 2; \quad \frac{d\theta}{dt} = \frac{2}{\sec^2 45^\circ} = 1$

10. **Ans (2)**

$\vec{A}\cdot\vec{B} = -8 - 5 + 15 = 2$

PART-2 : CHEMISTRY

SECTION-I (i)

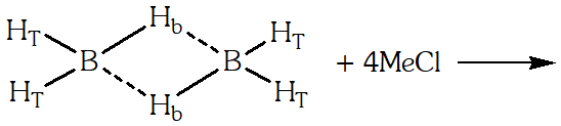
3. **Ans (B)**

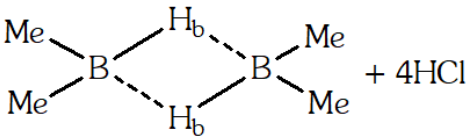
NH_3 sp_3 :107°

$(CH_3)_3N$ sp^3 : 108°-bulky have more repulsion than smaller H

$(SiH_3)_3N$ sp^2 : 120° LP of N engages in back bonding with empty d orbitals of Si

4. **Ans (A)**





Bond length longer due to e^- delocalise and stronger because of 3 Nucleus attraction.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C)**

$2Na_3PO_4 + 3Ba(NO_3)_2 \rightarrow Ba_3(PO_4)_2 + 6NaNO_3$

Moles taken	0.2	0.5
Moles reacted	0.2	0.3
Moles left	0	0.2

Na_3PO_4 is limiting reactant.

$\therefore$ Moles of $Ba_3(PO_4)_2$ formed = 0.1

Molarity of $Ba_3(PO_4)_2 = \frac{0.1}{1} = 0.1M$

6. **Ans (A,B,C)**

(A) In elementary reaction stoichiometry of reactant is order w.r.t. that reactant.

(B) Zero order reaction Rate = k (A)°

$\text{Rate} = k = \frac{\text{mol}}{\text{L sec}}$

(C) For zero order reaction Rate = k

k depends on temperature.

(D) A zero-order reaction cannot be an elementary reaction, as stoichiometry cannot be zero.

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (200.00)**

$\Delta T_f = i\,K_f m = K_f, \frac{w_B \times 1000}{M \times w_A}$

$0.093 = 1.86 \times \frac{1}{M} \times \frac{1000}{100}$

$M = \frac{18.6}{0.093} = 200$

2. **Ans (2.00)**

$\frac{\Delta P}{P_A^0} = \frac{n_B}{n_A + n_B} = \frac{15/60}{\frac{18}{18} + \frac{15}{60}} = 0.2$

PART-2 : CHEMISTRY

SECTION-II (ii)

6. **Ans (9)**

Acidic oxides

NO_2, Cl_2O_7, CO_2, Cl_2O, Mn_2O_7, CrO_3, SO_2, N_2O_5, SO_3

Neutral oxide

NO,

Amphoteric oxides

Cr_2O_3 , Al_2O_3

7. **Ans (5)**

|Useful work|= ΔG

$\Delta S = 2$ kJ/mol-K

$\Delta H = 2(-100) - 60 - 3(-285)$

$\Delta H = -200 - 60 + 855$

$\Delta H = 595$ kJ/mol

$\Delta G = \Delta H - T\Delta S$

$\Delta G = 595 - 300 \times 2$

$\Delta G = -5$ kJ/mol, $|\Delta G| = 5$ kJ/mole

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8. **Ans (2)**

$\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{1}{6} = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = \frac{5}{6}$

$\Rightarrow T_2 = \frac{5}{6}T_1$ ....(1)

$\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{1}{3} = 1 - \frac{T_2 - 65}{T_1} \Rightarrow \frac{T_2 - 65}{T_1} = \frac{2}{3}$

$\Rightarrow T_2 - 65 = \frac{2}{3}T_1$ ....(2)

by equation (1) & (2)

$\frac{5}{6}T_1 - 65 = \frac{2}{3}T_1$

$T_1 = 390\ K = 117°C$

and $T_2 = \frac{5}{6}T_1 = 325K = 52°C$

$\therefore \quad \frac{T_1}{T_2} = \frac{117}{52} = 2.25$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

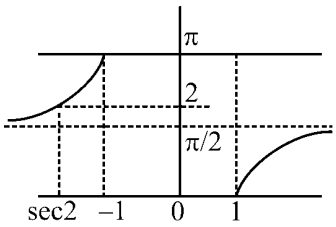
$9\cos^2 x + 16\sec^2 x$

$= 9[(\cos x - \sec x)^2 + 2] + 7\sec^2 x$

it is minimum when

$\cos x = \sec x = 1$

2. **Ans (C)**



$Df \longrightarrow (-\infty, \sec 2] \cup [1, \infty)$

3. **Ans (A)**

Using properties of A.P.,

we get $|S_1 - S_{101}| = 10$.

4. **Ans (A)**

$\Rightarrow \quad \frac{\alpha}{\beta} + \frac{\alpha}{\gamma} + \frac{\beta}{\alpha} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha} + \frac{\gamma}{\beta}$

$\Rightarrow \quad \frac{\alpha+\gamma}{\beta} + \frac{\alpha+\beta}{\gamma} + \frac{\beta+\gamma}{\alpha} = \frac{1-\beta}{\beta} + \frac{1-\gamma}{\gamma} + \frac{1-\alpha}{\alpha}$

$= -3 + \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = -3 + \frac{\frac{1}{2}}{-\frac{5}{2}}$

$= -3 - \frac{1}{5} = -\frac{16}{5}$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,C,D)**

$3f(x + 2y) = f(2x) + f(y) + 2x + 10y + 1\ \forall\ x, y \in R$

Let $x = y = 0 \Rightarrow f(0) = 1 \Rightarrow f'(0) = 2$

Differentiate (1) w.r.t. y

$6f'(x + 2y) = f'(y) + 10$

Now let $y = 0 \Rightarrow 6f'(x) = f'(0) + 10$

$\Rightarrow f'(x) = 2 \Rightarrow f(x) = 2x + 1 \ (\because f(0) = 1)$

6. **Ans (A,B,C,D)**

(A) $\cot x > \tan x\ ,\ \forall\ x \in \left(0, \frac{\pi}{4}\right)$

$\Rightarrow (\cot x)^{\ell n(\sin x)} < (\tan x)^{\ell n(\sin x)}$

$(\because\ \ell n(\sin x) < 0)$

(B) $4 < 5$

$\Rightarrow 4^{\ell n \operatorname{cosec} x} < 5^{\ell n \operatorname{cosec} x}$

(C) $\frac{1}{e} > \frac{1}{3}$

$\Rightarrow \left(\frac{1}{e}\right)^{\ell n \cos x} < \left(\frac{1}{3}\right)^{\ell n \cos x}$

(D) $\because\ \log_{1/\pi}(\tan x) < \log_{1/\pi}(\sin x)$

$\Rightarrow 2^{\log_{1/\pi}(\tan x)} < 2^{\log_{1/\pi}(\sin x)}$

7. **Ans (A,B,C,D)**

$(x^2 - x)^2 - 8(x^2 - x) + 12 < 0$

$(x^2 - x - 6)\ (x^2 - x - 2) < 0$

$(x - 3)\ (x + 2)\ (x - 2)\ (x + 1) < 0$

$x \in (-2, -1) \cup (2, 3)$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (25.00)**

$ar = 4 \quad \frac{a}{1 - r} = 25$

$r = \frac{1}{5}, \frac{4}{5}$ i.a = 20, a = 5.

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2. **Ans (3.00)**

$S_n = \dfrac{a(1-r^n)}{1-r}$

for $r = \dfrac{4}{5}$ & $a = 5$; for $r = \dfrac{1}{5}$ & $a = 20$

$\dfrac{5\left(1-\left(\frac{4}{5}\right)^n\right)}{1-\frac{4}{5}} > 24$

$\dfrac{20\left(1-\left(\frac{1}{5}\right)^n\right)}{1-\frac{1}{5}} > 24$

$\left(\dfrac{4}{5}\right)^n < \dfrac{1}{25}$ $\Rightarrow 1-\left(\dfrac{1}{5}\right)^n > \dfrac{24}{25}$

$n > \dfrac{2\log 5}{\log 5 - \log 4}$

$\Rightarrow \left(\dfrac{1}{5}\right)^n < \dfrac{1}{25} \Rightarrow 5^n > 25$

$n > \dfrac{2(1-0.3)}{1-0.3-2\times 0.3}$ $\Rightarrow n > 2 \Rightarrow n \geq 3$

$n > 14$

smallest value n = 15

3. **Ans (3.00)**

$f(x) = ax^3 + bx^2 + cx + d$

$f(1) = -1 \quad \therefore a + b + c + d = -1$ .... (i)

$f(2) = 18 \quad \therefore 8a + 4b + 2c + d = 18$ .... (ii)

$f''(0) = 0 \quad \therefore b = 0$ .... (iii)

$f'(-1) = 0 \quad \Rightarrow 3a - 2b + c = 0$ .... (iv)

Solving (i), (ii), (iii) and (iv)

$a = \dfrac{19}{4}, b = 0, c = \dfrac{-57}{4}, d = \dfrac{34}{4}$

$f(x) = \dfrac{1}{4}(19x^3 - 57x + 34)$

$f'(x) = \dfrac{1}{4}[57x^2 - 57] = 0$

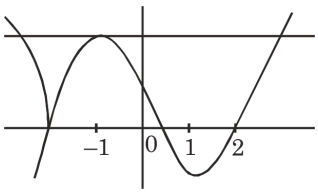
$x = 1, x = -1$

point of maxima $\equiv (-1, 18)$

point of minima = $(1, -1)$

Distance between the tangents is = 19 units

Also number of solutions of $|f(x)| - 18 = 0$



Three solutions.

4. **Ans (19.00)**

$f(x) = ax^3 + bx^2 + cx + d$

$f(1) = -1 \quad \therefore a + b + c + d = -1$ .... (i)

$f(2) = 18 \quad \therefore 8a + 4b + 2c + d = 18$ .... (ii)

$f''(0) = 0 \quad \therefore b = 0$ .... (iii)

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Solving (i), (ii), (iii) and (iv)

$a = \dfrac{19}{4}, b = 0, c = \dfrac{-57}{4}, d = \dfrac{34}{4}$

$f(x) = \dfrac{1}{4}(19x^3 - 57x + 34)$

$f'(x) = \dfrac{1}{4}[57x^2 - 57] = 0$

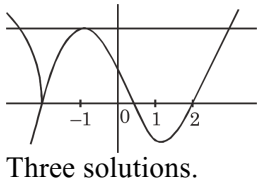
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Distance between the tangents is = 19 units

Also number of solutions of $|f(x)| - 18 = 0$



Three solutions.

PART-3 : MATHEMATICS

SECTION-II (ii)

5. **Ans (1)**

$f(x) = \dfrac{\alpha x}{x+1}$

$f(f(x)) = x$

$\alpha^2 = \alpha x + x + 1 \Rightarrow \alpha = -1$

$\alpha^2 = 1$

6. **Ans (9)**

$M = \lim\limits_{x\to 0}\left(\dfrac{1-(f(x))^3}{5\tan^2 x}\right)$

$= \lim\limits_{x\to 0}\left(\dfrac{(1-f(x))\left(1+f(x)+f^2(x)\right)}{5\left(\frac{\tan^2 x}{x^2}\right)x^2}\right)$

$\lim\limits_{x\to 0}\left(\dfrac{3(1-f(x))}{5x^2}\right)$

$= \dfrac{3}{5}\lim\limits_{x\to 0}\left(\dfrac{1-\cos 2x\cos 4x\cos 6x\cos 8x\cos 10x}{x^2}\right)$

$= \dfrac{3}{5}\left(\dfrac{2^2+4^2+6^2+8^2+10^2}{2}\right)$

$= \dfrac{3}{5}\times 2^2\left(\dfrac{1^2+2^2+3^2+4^2+5^2}{2}\right)$

$= \dfrac{12}{5}\times\dfrac{5.6.11}{12} = 66$

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7. **Ans (2)**

$f(x) = \max\left(|x|, \left|\frac{x}{3} - 1\right|\right)$



2 points when $f(x)$ is non-derivable

8. **Ans (40)**

$x^2 - x - 7 = 0$

α is a root of equation

$\alpha^2 - \alpha - 7 = 0 \Rightarrow \alpha - 1 = 7/\alpha$

similarly $(\beta - 1) = 7/\beta$

$\because f(n) = \alpha^n + \beta^n$

$$\frac{f(5) - f(4) + f(3)}{3f(3)}$$

$$= \frac{(\alpha^5 + \beta^5) - (\alpha^4 + \beta^4) + (\alpha^3 + \beta^3)}{3(\alpha^3 + \beta^3)}$$

$$= \frac{\alpha^4(\alpha - 1) + \beta^4(\beta - 1)}{3(\alpha^3 + \beta^3)} + \frac{1}{3}$$

$$\frac{\alpha^4(7/\alpha) + \beta^4(7/\beta)}{3(\alpha^3 + \beta^3)} + \frac{1}{3}$$

$$= \frac{7(\alpha^3 + \beta^3)}{3(\alpha^3 + \beta^3)} + \frac{1}{3} = \frac{8}{3}$$

9. **Ans (2)**

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{dx/dy}\right) = \frac{d}{dy}\left(\frac{1}{dx/dy}\right)\cdot\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{-1}{(dx/dy)^2}\frac{d^2x}{dy^2}\frac{1}{(dx/dy)}$$

$$\frac{d^2y}{dx^2} = \frac{-d^2x/dy^2}{(dx/dy)^3}$$

$y = \sin^2 x + \cos x$

$$\frac{dy}{dx} = \sin 2x - \sin x, \quad \frac{d^2y}{dx^2} = 2\cos 2x - \cos x$$

$$\frac{d^2x}{dy^2} = -\frac{(2\cos 2x - \cos x)}{(\sin 2x - \sin x)^3}$$

$$\left|\frac{d^2x}{dy^2}\right|_{x=\frac{\pi}{4}} = \frac{\frac{1}{\sqrt{2}}}{\left(1 - \frac{1}{\sqrt{2}}\right)^3} = \frac{\frac{1}{\sqrt{2}}.2\sqrt{2}}{(\sqrt{2} - 1)^3}$$

$= 2[2\sqrt{2} + 6 + 3\sqrt{2} + 1] = 14 + 10\sqrt{2}$

$\therefore a = 14, b = 10$

$$\therefore \frac{a + b + 2}{13} = \frac{26}{13} = 2$$

10. **Ans (7)**

Let C, H, F denote the sets of members who are on the cricket team, hockey team and football team respectively.

Then we are given $n(C) = 21, n(H) = 26, n(F) = 29$

$n(H \cap C) = 15, \quad (H \cap F) = 15, \quad n(F \cap C) = 12$

$n'(C \cap H \cap F) = 8$

$-n(C \cap H) - n(H \cap F) - n(F \cap C) + n(C \cap H \cap F)$

Hence, $n(C \cup H \cup F) = (21 + 26 + 29) - (14 + 15 + 12) + 8 = 43$

$4 + 3 = 7$

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